# Mohiuddin Bilwal

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LinkedIn | Portfolio | GitHub | Medium | ResearchGate | Google Scholar

## RESEARCH INTERESTS

High-Performance Computing, Model Compression Techniques, Machine and Deep Learning, Computer Vision, Large Language Models, Agentic Systems, Human-Computer Interaction, Software Engineering, Robotics

## EDUCATION

**BSc in Computer Science and Engineering** Jan 2020 – Apr 2024

Military Institute of Science and Technology, Dhaka, Bangladesh

Thesis: "Simulating User Behavior to Evaluate the Navigability of Websites Through Artificial Intelligence"

CGPA: 3.41 / 4.00

## TEST SCORE

**IELTS Academic:** 6.5 (Listening: 7.0, Reading: 6.5, Writing: 6.5, Speaking: 6.5) Oct 2026

## RESEARCH EXPERIENCE

**Researcher** Aug 2025 – Jul 2026

SwAPP Lab, Iowa State University | Supervisor: Prof. Ali Jannesari

- Performed structured pruning on CNN and ViT architectures for ProfilingAgent research, cutting memory by up to 7.7% while keeping accuracy loss below 1%.
- Reproduced structured pruning and quantization on CNN and ViT architectures (GETA), achieving up to 61% parameter and 44% MACs reduction while matching published baselines within 0.02%.
- Reviewed literature, built result tables, and fine-tuned models for an LLM-oriented model compression study.

**Research Assistant** Aug 2026 – Present

Elite Research Lab | Mentor: Prof. Kishor Morol

- Evaluated structured pruning against backdoor and weight watermarks on 5 architectures and 2 datasets, showing that removing 12% of parameters dropped backdoor detection from 99% to 24% while weight watermarks survived.
- Built an LLM agent that compressed ViT-B/16 while keeping the watermark verifiable in 7 of 8 searches, against 2 of 8 without watermark measurements.
- Developed a method to verify watermarks in pruned models using only the original model, with no help from the compressor.

## PUBLICATIONS

[1] S. Jafari, **M. Bilwal**, F. Zhou, B. Gelder, A. Jannesari, "APQF: Agentic Profiling-Guided Structured Pruning and Mixed-Precision Quantization with Adaptive Fine-Tuning.", International Conference for High Performance Computing, Networking, Storage, and Analysis (SC26) (Poster Accepted) (Under Review at MLSYS 2027). [preprint]

[2] S. Jafari, A. Sarkar, **M. Bilwal**, A. Jannesari, "ProfilingAgent: Profiling-Guided Agentic Reasoning for Adaptive Model Optimization," IEEE Computational Intelligence Society (Under Review). [preprint]

[3] **M. Bilwal**, S. Abrar, Z. Sultana, J. U. Tanvin, R. M. R. Shahriar, M. M. Rahman, M. N. Islam, "Proposing and Evaluating a Modified Keyboard Layout for One-Handed Users." 27th International Conference on Computer and Information Technology (ICCIT 2024), Cox's Bazar, Bangladesh. [ref]

[4] **M. Bilwal**, J. U. Tanvin, S. M. S. Rahman, M. R. F. Faisal, A. F. Khan, S. E. Arafat, M. Akhtaruzzaman, "A Multi-Modal Assistive Robot Navigation System for Physically Impaired Users." IEEE International Conference on Quantum Photonics, Artificial Intelligence, and Networking (QPAIN 2025), Bangladesh Army University of Science and Technology (BAUST), Saidpur, Bangladesh. [ref]

[5] A. Hossain, **M. Bilwal**, A. Sircar, M. N. Islam, "Simulating User Behavior to Evaluate the Navigability of Websites Through Artificial Intelligence." In Preparation, 2026. [ref]

[6] F. N. Nizami, **M. Bilwal**, A. Sircar, B. Ghosh, M. Akhtaruzzaman, S. R. Khan, N. N. Khanom, "Detection of Coronavirus Using CBC Blood Information." Draft. [ref]

## TECHNICAL SKILLS

**Programming:** Python, Java, Ruby, Golang, C++, ReactJS, Rails, SQL, Redis, REST API, FAST API
**ML / NLP:** PyTorch, TensorFlow, Hugging Face, spaCy, NLTK, RAG, LLM, scikit-learn, CNN, Vision Transformers
**Tools:** Git, GitHub, Docker, Gateway, POSTMAN, LaTeX, Linux, Microsoft Azure, AWS S3 Bucket
**Embedded & IoT:** Raspberry Pi, Arduino, GPU
**Other Knowledge:** Background Jobs, Gateways, Scheduler, Slurm, Pruning, Quantization, Distillation, OpenRouter API

## PROJECTS

**NextGen Wheelchair: A Semi-Autonomous Solution for Disabled People Using Head Motion** [Link]
Tech Tools: OpenCV, Machine Learning, Bootstrap, JavaScript, JSON

- Developed and manufactured a wheelchair prototype for individuals with mobility issues to drive independently.
- Integrated head motion, voice, joystick, and sonar-sensor-based control and alert systems as key features.
- Implemented a web platform for caregivers to manually operate the prototype via the web interface.

**CVCBC Predictor: Detection of Coronavirus Using CBC Blood Information of Patients** [Link]
Tech Tools: Machine Learning

- Investigated the advantages of using Complete Blood Count (CBC) alongside PCR in COVID-19 diagnostics by analyzing CBC data of 4474 patients in Brazil.

**Air Quality Prediction of Nairobi**
Tech Tools: Machine Learning, Autoregressive Model, MongoDB

- Worked on Africa's largest open data platform, "openAfrica" in WorldQuant University's virtual Data Science Lab.
- Prepared time series data and built an autoregressive model to predict PM 2.5 readings throughout the day.
- Tuned hyperparameters to improve model performance.

## ACHIEVEMENTS

1. Selected for the AWS AI & ML Scholars Future AWS Agent Engineer Nanodegree Program, Sponsored by AWS.
2. Completed the AI-Integrated Leadership Program (AILP), Cohort 01, 2026, offered by Aspire Institute. [Link]
3. Completed the prestigious Aspire Leaders Program 2025 (Cohort 02), funded by Aspire Institute, founded at Harvard University. [Link]
4. Achieved 3rd position in the internal Hackathon Competition at Barikoi Maps, 2025.
5. Participated in the National Collegiate Programming Contest (NCPC), Jahangirnagar University, 2024.
6. Achieved 24th position out of 70 teams in the ML Olympiad - CO2 Emissions Prediction Challenge, Kaggle, 2024.
7. Participated in the International Collegiate Programming Contest (ICPC), Asia Dhaka Regional, BUBT, 2023.
8. Achieved 34th position out of 100+ teams in LEETCON (Cybersecurity Competition), MIST, 2023.
9. Participated in the Inter-University Programming Contest (IUPC), SUST, 2023.
10. Two abstracts [Link] were accepted in the IEEE CS BDC Summer Symposium, Islamic University, Kushtia, 2023.
11. Achieved 1st position out of 36 universities worldwide in the University Rover Challenge (URC) 2021.

## TEACHING EXPERIENCE

**Mentor, Team Coordinator — MIST Computer Club, MIST Innovation Club** Jan 2021 - Mar 2023
Military Institute of Science and Technology (MIST)

- Conducted sessions on different algorithms and competitive programming problems.
- Mentored junior programmers and assisted in problem-solving sessions.

- Conducted workshops on basic web development languages to give newly admitted students an overview of web development.

## PROFESSIONAL EXPERIENCE

**Junior Software Engineer — Research & Development Department** Jul 2025 - Present

TechnoNext Software Limited, Dhaka, Bangladesh

- Lead backend engineering for SaaS products, including a multi-tenant project management platform.
- Scaled the platform from crashing at ~200 concurrent users to sustaining 2,000+ users on the same hardware, achieving a 10× increase in capacity without infrastructure upgrades.
- Optimized the platform's slowest endpoint using compound database indexes, query optimization, and Redis caching, significantly reducing response time and memory pressure.
- Tuned PostgreSQL memory configuration and connection pooling, and implemented rate limiting and request timeouts to improve stability during traffic spikes.
- Designed a four-layer authorization model spanning gateway, user, role, and module permissions to enforce strict multi-tenant access isolation.
- Owned end-to-end backend development, including architecture, code reviews, CI/CD, automated testing, and system design.
- Built an application-controlled CI/CD module with repository integration, pipeline tracking, deployment management, approval workflows, and notifications through Discord, Telegram, and Slack.

**Tech Stack:** Ruby on Rails, PostgreSQL, Redis, React.js, TypeScript, Node.js, REST API, Docker, Gateway

**Software Engineer Intern** May 2025 - Jul 2025

Barikoi Maps, Dhaka, Bangladesh

- Solved algorithmic challenges and applied competitive programming skills to optimize geospatial mapping solutions.
- Built tools, scripts, and internal APIs to automate and support development workflows.
- Worked with data structures, parsing systems, and backend logic.
- Collaborated with senior engineers and participated in code reviews.
- Learned best practices in software development, testing, and deployment.

**Tech Stack:** Python, PostgreSQL, FAST API, JavaScript, Networking

## CERTIFICATIONS

1. Udacity: Introducing Generative AI with AWS. [Link]
2. Microsoft Certified: Azure AI Fundamentals [Link]
3. LinkedIn: Project Management Foundations [Link]

## VOLUNTEER EXPERIENCE

1. Volunteered in the Event Management Committee at the National Collegiate Programming Contest (NCPC) 2020, organizing logistics including decorations, food distribution, and information dissemination for participating teams.
2. Organized events in the Campus Hour program, including Treasure Hunt, on behalf of MIST Computer Club (MCC).
3. Worked with the Event Management Committee to organize the Intra-Department Short Pitch Cricket and Football Tournament, 2024.
4. Worked as an Open Source Contributor in GirlScript Summer of Code, 2023.

## REFERENCES

Available upon request.